

Experiment Comparison between VGGNet and ResNet-50 on 3-class CIFAR-10 dataset

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During the experiment for the final report, two complete experiment logs were compared: one for **VGGNet** and one for **ResNet-50**, both trained on the exact same 3-class CIFAR-10 subset (*with categories of plane, car, bird*) containing 15,000 training images and 3,000 test images. As a summary of the experimental results, the following findings were observed.

Model	Epochs	Accuracy (%)	Avg. Train Time / Epoch (s)
VGG	20	≈ 99.4	~ 8
ResNet	15	$\approx 89-90$	~ 51

Table 1: Comparison of VGG-16 and ResNet-50 on CIFAR-10 (3 classes).

1. Result Observation

Upon observing the results, it was found that VGG achieved higher accuracy ($\sim 99\%$) than ResNet ($\sim 89\%$) on this small, low-resolution, 3-class dataset of CIFAR-10. This was an unexpected result considering the theoretical advantages of ResNet.

Theoretically, ResNet is supposed to outperform VGG because of its core concept of **residual learning**. In ResNet, skip connections allow each block to learn only the residual $F(x)$, which represents the difference between input and output. This design has three major theoretical advantages: (1) Prevents vanishing gradients in very deep networks, (2) allows hundreds or thousands of layers without degradation in training accuracy, enables identity mapping at initialization, (3) allowing the network to behave like a shallow model and progressively learn deeper representations.

Due to these advantages, ResNet typically achieves far superior performance compared to VGG-16 on large, complex datasets such as ImageNet.

2. Analysis of Experimental Results

However, in our experiment using the CIFAR-10 subset (3 classes only), VGG almost fits the data perfectly due to the simplicity of the task. Adding 34 more layers to form ResNet-50 introduces unnecessary capacity, which may even harm generalization.

Furthermore, ResNet-50 was originally designed for high-resolution ImageNet images. After multiple stride-2 downsamplings on 32×32 CIFAR-10 images, the spatial resolution becomes extremely small, causing loss of spatial detail. VGG, on the other hand, uses smaller strides and pooling layers, thus preserving more spatial information at small scales.

Another contributing factor is the training schedule. Deep architectures such as ResNet require longer training periods and learning-rate decay strategies (e.g., step or cosine scheduling) to converge effectively. In this experiment, both models were trained with a constant learning rate for only 15–20 epochs. Under these constraints, ResNet had not yet fully converged.

3. Discussion

The above accuracies (around 99% for VGG and approximately $89-90\%$ for ResNet) were achieved with the combination of crop and flip, without setting the final BatchNorm’s gamma value to 0.

3.1. Effect of Data Augmentation

With `RandomHorizontalFlip` only, each data sample has a **50%** chance to be flipped left or right, making the model robust to orientation. This usually improves test accuracy slightly because, theoretically, the model sees more variety. With `RandomCrop`, the same training image might appear shifted a little in different directions every epoch, leading to a stronger improvement in theory.

To analyze the influence of data augmentation, two augmentation settings (combination of `RandomHorizontalFlip` + `RandomCrop` and `RandomCrop` only) were tested on the same ResNet-50 network.

When trained with only `RandomCrop`, VGGNet reached an unexpected high test accuracy of **98.8%**, while the combination achieved around **89%**. This large gap for VGGNet was initially surprising because, in theory, the combination of crop and flip should outperform the use of a single augmentation. However, VGGNet’s high sensitivity to spatial orientation caused it to struggle with the added variation from horizontal flipping. If the dataset contained directionally biased images, flipped samples introduced unrealistic patterns that confused the model. As a result, combining crop and flip augmentations reduced generalization performance compared to using `RandomCrop` alone. For ResNet, crop augmentation alone gave the accuracy of **90.8%** while the combination gave **99.99%**, aligning by the theoretical claim.

3.2. Effect of Setting BatchNorm Gamma to Zero

In each ResNet bottleneck block, the residual branch ends with a 1×1 convolution followed by Batch Normalization (BN). Let the BN scale parameter be γ (channel-wise). Setting the *final* BN’s γ to zero initializes the residual branch to output approximately zeros, so the block computes

$$y = x + F(x; \theta) \approx x + 0 = x \quad (\text{identity mapping at initialization}).$$

This *identity initialization* stabilizes early optimization by preventing large and noisy residual updates at the beginning of training, while still allowing the residual path to grow as γ learns away from zero.

Experimental Setup. We compare two variants on the same ResNet-50 configuration and the same fixed CIFAR-10 (3-class) train/test split:

- **Default BN:** Standard initialization (PyTorch defaults) for all BN layers.
- **Identity Init (BN $\gamma=0$):** For each bottleneck block, set $\gamma=0$ in the *last* BN (the BN following the final 1×1 convolution) before training starts.

Although identity initialization (setting the final BN’s γ to 0) is commonly used in ResNet architectures, its effect was also tested on VGGNet for comparison. The results showed an interesting contrast between the two architectures. For VGGNet, the default BN initialization ($\gamma=1$) led to a final test accuracy of about **89%**, whereas setting $\gamma=0$ boosted performance dramatically to around **99.6%**. Training with $\gamma=0$ produced smoother convergence, lower initial loss, and more stable accuracy progression across epochs.

In contrast, applying the same identity initialization to ResNet caused a performance drop—from approximately **99%** to **87%**. This degradation occurs because ResNet already includes explicit identity skip connections. By forcing the residual branch to start with zero scaling, the network effectively behaved as a pure identity mapping during early training, preventing sufficient residual learning and causing slower convergence and mild underfitting.

Empirically, initializing the final BN scale to zero tends to:

- **Stabilize early training:** Accuracy rises smoothly in the first few epochs, as the network begins close to an identity map and avoids large residual perturbations.
- **Yield comparable or slightly higher final accuracy:** On small, low-variance datasets (e.g., 3-class CIFAR-10), the difference is often modest because the task is already easy and the networks are overparameterized.

In more challenging scenarios (more classes, deeper networks, or stronger regularization), identity initialization can yield clearer benefits by improving optimization stability and gradient flow across many stacked residual blocks. In our experiments, its main observable effect was *smoother convergence*, while the final accuracy varied depending on the architecture.

4. Conclusion

In summary, although ResNet-50 is theoretically more powerful than VGG-16, the results demonstrate that **architecture superiority is dataset-dependent**. On small, low-resolution, and relatively simple datasets such as the 3-class subset of CIFAR-10, VGG’s straightforward convolutional design combined with batch normalization can achieve higher accuracy than deeper residual networks. In such settings, model depth and residual complexity may not translate into better performance, as the task does not fully exploit their representational capacity.

The experiments on Batch Normalization initialization further reveal that architectural behavior is highly sensitive to parameter initialization. For VGGNet, setting the final BatchNorm’s γ value to zero significantly improved both convergence speed and final test accuracy (from **89%** to **99.6%**), showing that identity initialization can act as a strong stabilizer in non-residual architectures. In contrast, applying the same initialization to ResNet-50 led to a noticeable performance drop (from **99%** to **87%**), since its residual design already includes an implicit identity mapping. This indicates that the effectiveness of identity initialization depends on the presence of skip connections and the inherent stability of the network’s topology.

Overall, the findings emphasize that initialization strategies and architectural choices must be aligned with dataset scale and complexity. While ResNet architectures are optimized for deep hierarchical representations and large-scale tasks, simpler networks like VGG can be more efficient and even superior in smaller domains. In addition, fine-tuning BatchNorm parameters—particularly the final scale factor γ —can play a surprisingly large role in optimization dynamics.